2.1 Solutions and Solution Sets

For each of the following determine if the given number is a solution to the given equation or inequality.

1. Is $x = 6$ a solution to $2x - 5 = 3(1 - x) + 22$?
2. Is $t = 7$ a solution to $t^2 + 3t - 10 = 4 + 8t$?
3. Is $t = -3$ a solution to $t^2 + 3t - 10 = 4 + 8t$?
4. Is $w = -2$ a solution to $\dfrac{w^2 + 8w + 12}{w + 2} = 0$?
5. Is $z = 4$ a solution to $6z - z^2 \geq z^2 + 3$?
6. Is $y = 0$ a solution to $2(y + 7) - 1 < 4(y + 1) + 3(4y + 10)$?
7. Is $x = 1$ a solution to $(x + 1)^2 > 3x + 1$?

2.2 Linear Equations

Solve each of the following equations and check your answer.

1. $4x - 7(2 - x) = 3x + 2$
2. $2(w + 3) - 10 = 6(32 - 3w)$
3. $\dfrac{4 - 2z}{3} = \dfrac{3}{4} - \dfrac{5z}{6}$
4. $\dfrac{4t}{t^2 - 25} = \dfrac{1}{5 - t}$
5. $\dfrac{3y + 4}{y - 1} = 2 + \dfrac{7}{y - 1}$
6. $\dfrac{5x}{3x - 3} + \dfrac{6}{x + 2} = \dfrac{5}{3}$

2.11 Linear Inequalities

For problems 1 - 6 solve each of the following inequalities. Give the solution in both inequality and interval notations.

1. $4(z+2)-1>5-7(4-z)$
2. $\frac{1}{2}(3+4t)\le 6\left(\frac{1}{3}-\frac{1}{2}t\right)-\frac{1}{4}(2+10t)$
3. $-1<4x+2<10$
4. $8\le 3-5z<12$
5. $0\le 10w-15\le 23$
6. $2<\frac{1}{6}-\frac{1}{2}x\le 4$
7. If $0\le x<3$ determine a and b for the inequality : $a\le 4x+1<b$